

Name: _____

Date: _____

2D Projectile Motion: Launch At Arbitrary Angles

SYNOPSIS

A projectile launched at an angle is really two separate motions running on one clock: a horizontal motion at constant velocity, and a vertical motion in freefall. They don't interact — gravity never touches the horizontal component, and nothing about the horizontal component changes how fast the projectile falls. Splitting V_0 into $V_{0x}=V_0\cos\theta$ and $V_{0y}=V_0\sin\theta$ lets us reuse the freefall equations from Note 3 on the vertical component, and constant-velocity equations on the horizontal one. Because only the vertical component ever stops and reverses, the flight time depends only on V_{0y} — the horizontal speed is along for the ride.

PERFORMANCE EXPECTATIONS

Resolve a launch velocity into horizontal and vertical components, use the independence of those components to find flight time and maximum height, and work a complete numeric launch problem including range and impact velocity.

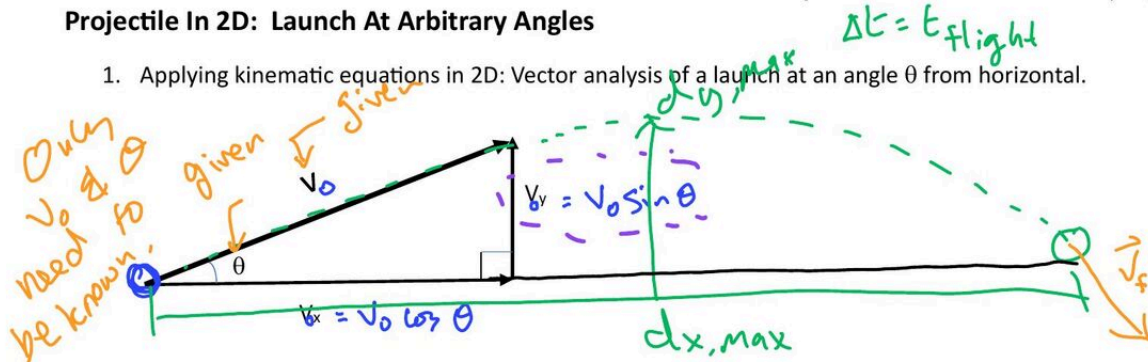
Instructions and Questions

1. A projectile launches with speed V_0 at angle θ above horizontal. Write the formulas for the horizontal and vertical components of the launch velocity:

$V_{0x} =$ _____ $V_{0y} =$ _____

Projectile In 2D: Launch At Arbitrary Angles

1. Applying kinematic equations in 2D: Vector analysis of a launch at an angle θ from horizontal.



1) How long does the projectile stay in the air?
Flight time Δt (abbr. as just t)

Note: Flight time ONLY depends on V_{0y}
ie vertical component! And horizontal motion
lasts as long as in flight.

$$V_y = V_{0y} - 9.8t = V_0 \sin \theta - 9.8t$$

Note: what is the V_y at apex? ZERO !!
what is the V_y when falls back (impd)
 $-V_{0y}$

$$\text{at Apex } V_y = 0 = V_{0y} - 9.8t_{1/2}$$

$$\text{at impact } V_y = -V_{0y} = V_{0y} - 9.8t_{\text{fall}}$$

Figure 1 — component setup and flight-time derivation, as drawn by Dr. Cho.

2. At the very top of the flight path — the **apex** — what is V_y at that instant? _____

In your own words, why does flight time depend only on V_{0y} and not on V_{0x} ?

3. Derive t_{flight} for a launch and landing at the *same* height. Start from $V_y = V_{0y} - gt$, and use the fact that the projectile returns to launch height moving at $-V_{0y}$. Show your steps:

$$t_{\text{flight}} = \underline{\hspace{2cm}}$$

4. Derive the max-height formula. Start from $v_y^2 = v_{0y}^2 - 2g \cdot d$, and use $v_y = 0$ at the apex. Show your steps:

$$d_{y,\text{max}} = \underline{\hspace{2cm}}$$

5. Worked example: a ball launches at $V_0 = 10 \text{ m/s}$, 25° above horizontal, from the edge of a 1.5 m cliff. Use $g = 9.8 \text{ m/s}^2$ throughout.

- a) Components: $V_{0x} = \underline{\hspace{1cm}} \text{ m/s}$ $V_{0y} = \underline{\hspace{1cm}} \text{ m/s}$

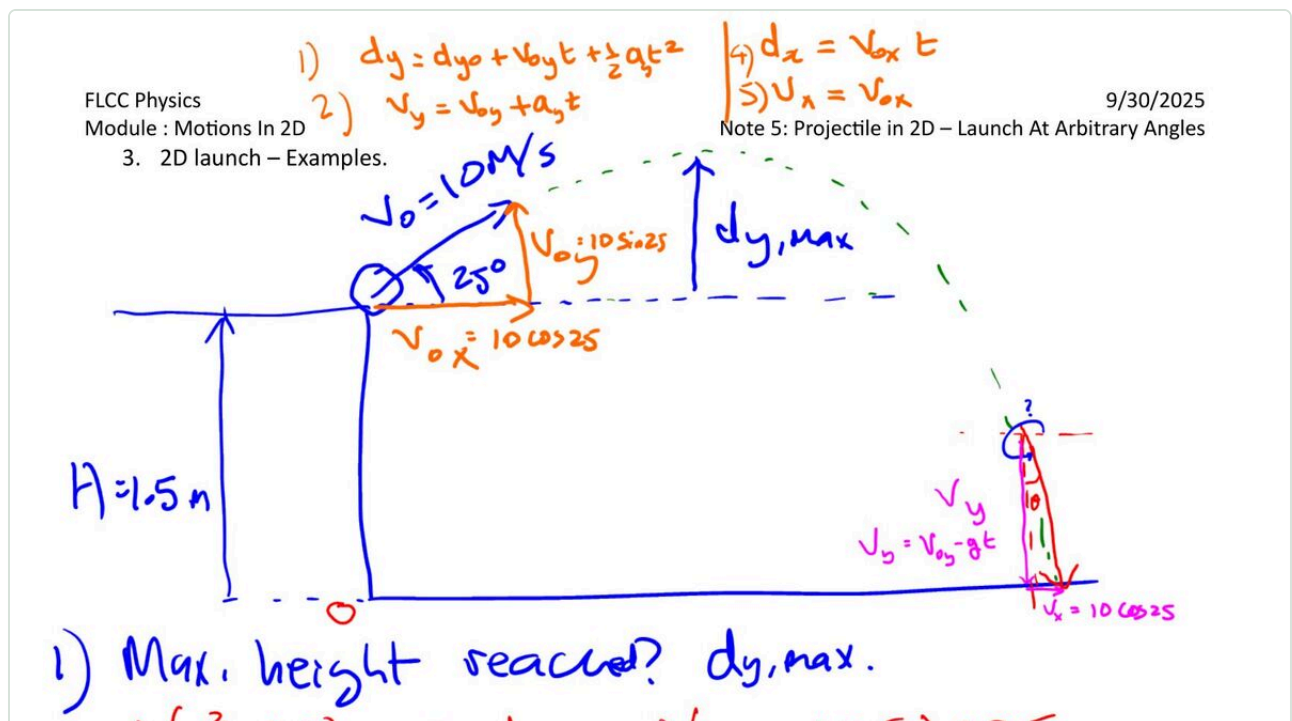


Figure 2 — the worked-example setup, as drawn by Dr. Cho.

- b) Max height above the launch point: $d_{y,\text{max}} = \underline{\hspace{1cm}} \text{ m}$

- c) This is a cliff launch, so launch and landing heights differ. Write the vertical position equation with the landing point 1.5 m *below* the launch point, and solve the resulting quadratic for t :

$$t_{\text{flight}} = \underline{\hspace{2cm}} \text{ s}$$

d) Range: $d_{x,\text{max}} = \underline{\hspace{2cm}} \text{ m}$

e) Impact velocity: $V_x = \underline{\hspace{2cm}} \text{ m/s}$ $V_y = \underline{\hspace{2cm}} \text{ m/s}$ speed = $\underline{\hspace{2cm}} \text{ m/s}$

6. General cliff-launch setup: starting from $d_x = (V_0 \cos \theta)t = D$ (solve for t) and the vertical position equation with landing height $-H$ relative to launch, eliminate t and use $1/\cos^2 \theta = 1 + \tan^2 \theta$ to show the landing condition is a quadratic equation. Let $x = \tan \theta$ and write the boxed general form:

$$0 = \underline{\hspace{2cm}} + D \cdot x - \underline{\hspace{2cm}} \cdot x^2$$

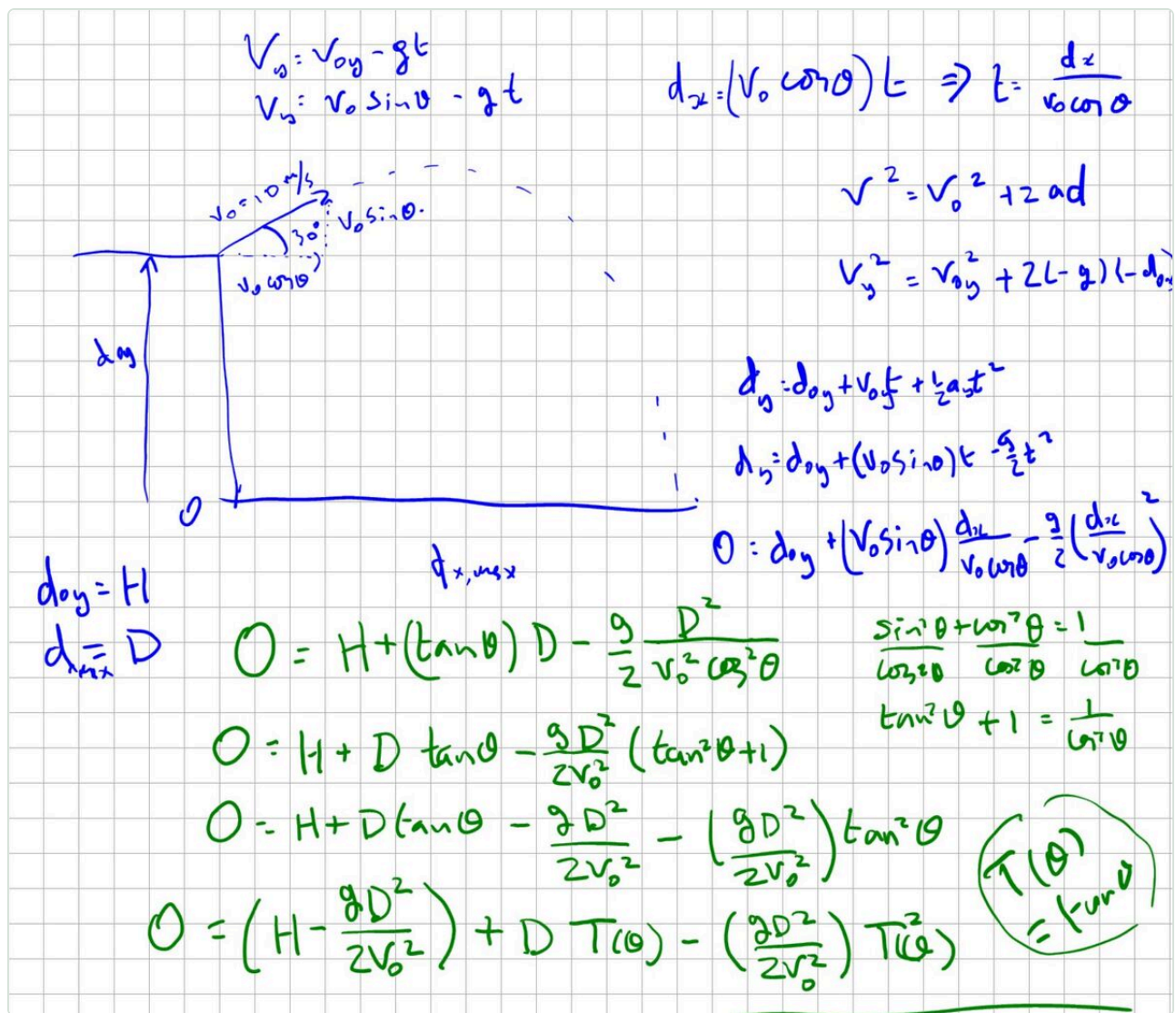


Figure 3 — the general cliff-launch derivation, as drawn by Dr. Cho.

7. A quadratic in $\tan \theta$ generally has two roots. What do the two solutions represent, physically, for a launch that has to clear distance D from height H ?